

# AI-Powered MPLADS Audit Intelligence Assistant

A Unified Spatio-Temporal, Multi-Agent Anomaly Detection & Asset Evidence Forensics System for Public Infrastructure Auditing

💡 **EXECUTIVE PITCH IN ONE SIMPLE SENTENCE**

"Instead of an auditor manually opening 1,00,000 opaque project records, our AI scans all MPLADS data, uncovers hidden financial, spatial, contractor, and photo anomalies, gives each project an explainable 0–100 priority score, and tells human auditors exactly where and why to investigate."

## 1. Problem Context & The Operational Reality

Under the **MPLADS** scheme, crores of public funds are sanctioned across constituencies for local development works. Every project passes through 5 stages: **Recommended** → **Sanctioned** → **Started** → **Payments** → **Completed**. Portals like **eSAKSHI** and **data.gov.in** catalog these records. However, because audit officers face **millions of rows**, manual examination is impossible. Duplicate tenders, contractor cartels, and inflated costs slip past traditional siloed spreadsheets unnoticed.

| Work Name         | Cost    | Location / Cadastral         | Contractor | Status    | What Our AI Flags                                                             |
|-------------------|---------|------------------------------|------------|-----------|-------------------------------------------------------------------------------|
| Community Hall    | ₹20.0 L | Village A (22.3001, 70.8002) | ABC Ltd    | Completed | Standard Baseline Reference                                                   |
| Village Road      | ₹18.0 L | Village B                    | XYZ Ltd    | Completed | Standard Baseline Reference                                                   |
| Community Hall    | ₹20.0 L | Village A (22.3003, 70.8001) | ABC Ltd    | Completed | ⚠️ <b>Duplicate Alert:</b> Identical cost, identical site (<50m), same vendor |
| Road Construction | ₹17.5 L | Village C                    | XYZ Ltd    | Ongoing   | ⚠️ <b>Cartel Risk:</b> Contractor XYZ monopolizes 80% tenders                 |

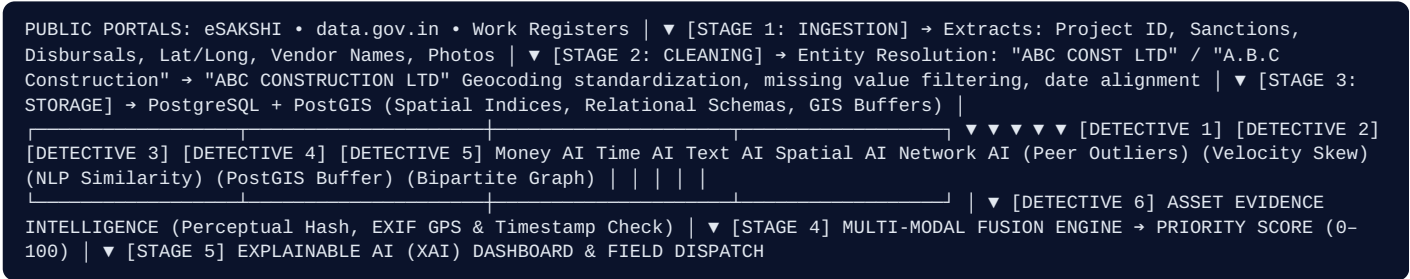
❌ **What our AI NEVER does:**

It does NOT pronounce legal guilt or say *"This officer/contractor is corrupt."* Autonomous accusations without physical ground truth violate the Indian Evidence Act and are legally indefensible.

✅ **What our AI ACTUALLY does:**

It provides an *advisory priority index* with corroborating forensic proof (maps, visual overlaps, and time inversions), pointing the human auditor directly to what requires physical ground verification.

## 2. End-to-End System Pipeline & Data Normalization



### 3. The 6 Multi-Modal "Detectives" (Specialized Analytical Engines)

Rather than an opaque single black-box algorithm, the platform deploys 6 specialized, interpretable detectives:

1. Money Detective (Financial Outliers)

Cost Anomaly

Calculates dynamic median costs for specific civil works in the same district and terrain. Flags works deviating sharply from local peer baselines.

Signal: Village road sanctioned at ₹35L vs district baseline of ₹11.2L (+212% standard deviation outlier).

2. Time Detective (Velocity & Inversion)

Temporal Anomaly

Tracks sequence milestones: Recommendation → Sanction → Disbursal → Completion. Flags disbursements released faster than physical construction timelines.

Signal: 100% of ₹48.5L budget disbursed within 48 hours of administrative sanction before foundation excavation.

3. Text Detective (NLP Duplicate Engine)

Semantic Match

Applies NLP and cosine text similarity on work descriptions, Bill of Quantities (BOQ), and specifications to detect duplicated scope of work.

Signal: 91.2% semantic BOQ overlap between two distinct tender allocations within the same municipal ward.

4. Location Detective (Spatial Clusters)

PostGIS Engine

Executes radial buffer queries using PostGIS. Identifies unnatural clustering where multiple high-outlay projects overlap within tiny micro-radii.

Signal: 5 separate solar lighting installations sanctioned inside an overlapping 140-meter cadastral buffer.

5. Network Detective (Graph Cartels)

Network Graph

Builds bipartite graph networks mapping relationships between agencies, contractors, sub-vendors, and sanctioning divisions.

Signal: Vendor "Apex Infra-Power" holds 88% of electrical awards across 2 adjacent wards via revolving sub-entities.

6. Evidence Detective (Image Forensics)

Computer Vision

Analyzes uploaded physical site photographs using Perceptual Hashing (pHash) and visual embeddings to detect cross-project asset reuse.

Signal: Uploaded completion photo matches historical project photo with 98.4% visual overlap and identical wall graffiti.

### 4. Asset Evidence Intelligence Deep-Dive & EXIF Validation

The Evidence Detective addresses the gap between digital entries and ground reality by analyzing image assets and metadata:

| Forensic Check            | How the AI Inspects the Evidence                                                                                           | Advisory Output & Safety Rule                                                                                                                     |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Exact Photo Reuse         | Computes SHA-256 cryptographic hashes and pHash signatures across all completion photos in the database.                   | Flags instant duplicate collision: "Identical photo uploaded for Project A & Project B."                                                          |
| Modified / Edited Images  | Employs visual feature embeddings (ResNet-50 / SSIM) resilient to cropping, resizing, lighting changes, or re-compression. | Flags visual similarity: "92% structural similarity — potential reused photographic evidence."                                                    |
| GPS Geotag Verification   | Extracts camera EXIF GPS coordinates and compares against official sanctioned cadastral boundary polygons.                 | If photo GPS points to Village B while record says Village A: "Location inconsistency detected." (If absent, notes "Unavailable"; never guesses). |
| Timestamp Chronology      | Cross-references EXIF capture timestamp against administrative sanction and completion sign-off dates.                     | Flags inversion: "Photo taken in Aug 2024, but work officially completed in Dec 2024."                                                            |
| Graceful Missing Fallback | If records have no images or stripped EXIF, the system switches dynamically to financial, spatial, and network signals.    | Marks status as "Insufficient Evidence" rather than hallucinating or producing false positives.                                                   |

#### The "Asset Fingerprint" Concept

- Comprehensive Identity Card for every project:**
- **What:** Scope, BOQ specifications, materials.
  - **Where:** PostGIS coordinates, cadastral ward.
  - **Cost:** Outlay, disbursement speed, tranches.
  - **Who:** Executing contractor, sub-vendors, nodal agency.
  - **Evidence:** Image hashes, EXIF telemetry, site logs.
- Enables instant multi-attribute querying across the database.

#### Connected Evidence Graph

MCD Div-7 Electrical | ▼ Apex Infra-Power Ltd |  
Project DL-8821  
(₹48.5L) Project DL-7102 (₹45.0L) Ward 42-B (Mehrauli)  
Ward 41-A (Lado Sarai) | |  
140m Proximity Buffer Overlap + 98.4% Photo Collision  
(pHash Match)

5. Auditor Executive Suite & Forensic UI Architecture

Designed specifically for state oversight bureaus and CAG auditors, providing intuitive evidence exploration:

| MPLADS Audit Intelligence Platform — Executive Forensic Queue |                                            |                         |                          |                                |                                                |                            |
|---------------------------------------------------------------|--------------------------------------------|-------------------------|--------------------------|--------------------------------|------------------------------------------------|----------------------------|
| 14,820                                                        | 142                                        | 8,940                   | 19                       | 94.2%                          |                                                |                            |
| Monitored Works                                               | High-Priority Cases                        | With Asset Photos       | Cross-Project Photo Dups | Data Quality Index             |                                                |                            |
| Project ID                                                    | Work Description & Nodal Agency            | Constituency            | Priority Score           | Corroborating Signals          | Evidence Forensic Status                       | Action                     |
| MP-2024-DL-8821                                               | Solar Street Lighting Phase IV (MCD Div-7) | South Delhi (Ward 42-B) | 91 / 100 (Critical)      | Financial + Spatial + Evidence | ⚠️ 98.4% Photo Collision + 140m Buffer Overlap | Immediate Field Inspection |
| MP-2024-MH-4402                                               | Community Skill Center Renovation (PWD)    | Pune Cantt, MH          | 84 / 100 (High)          | Temporal + Financial           | Disbursement velocity anomaly (100% in 4 days) | Document Verification      |
| MP-2024-KA-1193                                               | Borewell & Rural RO Water Plant (Zilla P.) | Mandya, KA              | 78 / 100 (High)          | Text Similarity + Network      | EXIF Stripped; Contractor cartel nexus         | Geo-tag Re-capture Order   |

6. Featured Case Study: Project #MP-2024-DL-8821

**CASE DOSSIER: MP-2024-DL-8821 — Solar Street Lighting Installation (Outlay: ₹48.50 Lakhs)**

Executing Vendor: Apex Infra-Power Ltd | Nodal Agency: MCD Div-7 Electrical | Priority: 91 / 100 (Critical Tier-1)

- Photo Duplication Collision:** Uploaded commissioning photo of solar pole demonstrates a **98.4% perceptual hash match** with archived project MP-2023-DL-7102 (commissioned 11 months earlier). Identical wall crack and municipal stencil visible.
- Cadastral Proximity Anomaly:** PostGIS spatial query reveals the two works share an overlapping **140-meter cadastral buffer** across Ward 42-B and Ward 41-A.
- Disbursement Velocity Skew:** 98.4% of total outlay (₹47.72 Lakhs) was disbursed in a single tranche within **48 hours** of award, breaching circular #FIN-08.
- Contractor Cartel Nexus:** Apex Infra-Power Ltd holds 88% of local electrical awards across MCD Div-7 under related registration variants.

**Recommended Field Action:** Dispatch junior auditor with tamper-resistant GPS camera app to verify engraved serial numbers on solar luminaires and confirm whether physical installation exists at (28.5244° N, 77.1855° E).

## 7. Strategic Pitching & Comparative Value for Hackathon Judges

| Evaluation Dimension      | Existing Legacy Portals (eSAKSHI)                              | Our AI Audit Intelligence Solution                                                         |
|---------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Operational Workflow      | Passive data warehouse: "Here are 1,00,000 unranked rows."     | Active forensic triage: "Here are the top 142 works needing physical audit today."         |
| Evidence Verification     | Manual photo review (auditors miss reused or cropped files).   | Automated pHash & ResNet visual embedding comparison across all historical projects.       |
| Spatio-Temporal Analysis  | Static pinpoints on simple maps.                               | PostGIS spatial buffer clusters + payment velocity timeline validation.                    |
| System Integration        | Would require immense expenditure to replace.                  | Non-invasive analytical overlay operating directly on existing databases.                  |
| CAG & Legal Defensibility | Spreadsheet-based ad-hoc checks vulnerable to human oversight. | Human-in-the-loop Explainable AI (XAI) strictly adhering to Indian Evidence Act standards. |

## 8. Legal Compliance, Implementation Blueprint & Scalability



### Statutory Defensibility (Indian Evidence Act):

- Every anomaly alert acts purely as an *evidentiary lead*.
- Hash collisions (SHA-256 / pHash) provide mathematically verifiable baselines.
- Signed auditor field logs create an unbroken chain of custody for CAG scrutiny.



### Enterprise Deployment & Scalability:

- **Database:** PostgreSQL with PostGIS extension for spatial queries.
- **Vision Pipeline:** GPU-accelerated embedding inference + Milvus vector search.
- **API Layer:** Read-only REST connectors to eSAKSHI & data.gov.in schemas.



### The Winning Presentation Summary:

"We do not ask the government to discard existing monitoring software like eSAKSHI. We add the missing forensic intelligence layer—connecting financial outlays, satellite GIS buffers, contractor cartels, and photographic evidence into a single explainable priority score that protects public funds."